

Euler–Lagrange derivations for the corrected SAT.O–Blockwave Lagrangian

We start from the corrected, dimensionally consistent Lagrangian density (Eq. (1) below) used in the main text:

$$\mathcal{L}(\rho, \mathbf{v}, \Phi) = \frac{1}{2}\rho\mathbf{v}^2 - \rho W(\Phi) - \frac{\kappa}{2}(\nabla\Phi)^2 + \lambda\rho\Phi(\nabla\cdot\mathbf{v}), \quad (1)$$

where $W(\Phi)$ is the potential per unit mass (units of Φ), κ is the curvature coupling, and λ is dimensionless. The independent fields for the variational principle are taken to be $\rho(\mathbf{x}, t)$, $\mathbf{v}(\mathbf{x}, t)$, $\Phi(\mathbf{x}, t)$. Mass conservation (the continuity equation) is imposed as a constraint using a Lagrange multiplier $\chi(\mathbf{x}, t)$. The constrained action is

$$S[\rho, \mathbf{v}, \Phi, \chi] = \int_{t_0}^{t_1} \int_{\Omega} \left\{ \mathcal{L}(\rho, \mathbf{v}, \Phi) + \chi(\partial_t \rho + \nabla \cdot (\rho \mathbf{v})) \right\} d^3x dt. \quad (2)$$

Below we compute the stationary conditions $\delta S = 0$ under arbitrary compactly-supported variations $\delta\rho$, $\delta\mathbf{v}$, $\delta\Phi$, $\delta\chi$. We integrate by parts and drop boundary terms (assuming either vanishing variations on the boundary or appropriate boundary conditions).

1. Variation with respect to the Lagrange multiplier χ .

Varying S with respect to χ enforces the continuity constraint:

$$\delta_\chi S = \int \delta\chi (\partial_t \rho + \nabla \cdot (\rho \mathbf{v})) d^3x dt \implies \boxed{\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0.}$$

2. Variation with respect to the velocity field $\mathbf{v}$.

Collect $\mathbf{v}$ -dependent pieces of the integrand:

$$\mathcal{L} \supset \frac{1}{2}\rho\mathbf{v}^2 + \lambda\rho\Phi(\nabla\cdot\mathbf{v}) + \chi\nabla\cdot(\rho\mathbf{v}).$$

Take the variation $\mathbf{v} \mapsto \mathbf{v} + \delta\mathbf{v}$. Compute term-by-term.

(i) Kinetic term:

$$\delta\left(\frac{1}{2}\rho\mathbf{v}^2\right) = \rho\mathbf{v} \cdot \delta\mathbf{v} \quad (\rho \text{ treated as independent}).$$

(ii) Interaction term:

$$\delta(\lambda\rho\Phi\nabla\cdot\mathbf{v}) = \lambda\rho\Phi\nabla\cdot(\delta\mathbf{v}) = -\lambda\nabla(\rho\Phi) \cdot \delta\mathbf{v} + (\text{boundary term}).$$

(We integrated by parts once and discarded the boundary term.)

(iii) Constraint term:

$$\delta(\chi \nabla \cdot (\rho \mathbf{v})) = \chi \nabla \cdot (\rho \delta \mathbf{v}) = -\rho \nabla \chi \cdot \delta \mathbf{v} + (\text{boundary term}).$$

Summing (i)–(iii) and requiring stationarity for arbitrary $\delta \mathbf{v}$ gives the Euler–Lagrange condition:

$$\boxed{\rho \mathbf{v} - \lambda \nabla(\rho \Phi) - \rho \nabla \chi = \mathbf{0} .}$$

Equivalently,

$$\boxed{\mathbf{v} = \nabla \chi + \frac{\lambda}{\rho} \nabla(\rho \Phi) .} \quad (3)$$

This equation expresses the velocity field in terms of the multiplier χ (a Bernoulli-type potential) and the interaction-gradient term. It is the canonical momentum relation coming from the variational principle.

3. Variation with respect to the density ρ .

Collect ρ -dependent terms from the integrand:

$$\mathcal{L} \supset \frac{1}{2} \rho \mathbf{v}^2 - \rho W(\Phi) + \lambda \rho \Phi \nabla \cdot \mathbf{v} + \chi (\partial_t \rho + \nabla \cdot (\rho \mathbf{v})).$$

Vary $\rho \mapsto \rho + \delta \rho$. Compute contributions (we treat $\mathbf{v}, \Phi, \chi$ fixed while varying ρ).

(i) From kinetic term:

$$\delta(\frac{1}{2} \rho \mathbf{v}^2) = \frac{1}{2} \mathbf{v}^2 \delta \rho.$$

(ii) From potential term:

$$\delta(-\rho W(\Phi)) = -W(\Phi) \delta \rho.$$

(iii) From interaction:

$$\delta(\lambda \rho \Phi \nabla \cdot \mathbf{v}) = \lambda \Phi (\nabla \cdot \mathbf{v}) \delta \rho.$$

(iv) From constraint term (two pieces):

$$\delta(\chi \partial_t \rho) = \chi \partial_t (\delta \rho) = -(\partial_t \chi) \delta \rho + (\text{boundary in time}),$$

and

$$\delta(\chi \nabla \cdot (\rho \mathbf{v})) = \chi \nabla \cdot (\delta \rho \mathbf{v}) = -\mathbf{v} \cdot \nabla \chi \delta \rho + (\text{spatial boundary}).$$

Combining all contributions and discarding total time/spatial boundary terms, stationarity with respect to $\delta \rho$ yields

$$\frac{1}{2} \mathbf{v}^2 - W(\Phi) + \lambda \Phi (\nabla \cdot \mathbf{v}) - \partial_t \chi - \mathbf{v} \cdot \nabla \chi = 0.$$

Rearrange to the more suggestive form

$$\boxed{\partial_t \chi + \mathbf{v} \cdot \nabla \chi = \frac{1}{2} \mathbf{v}^2 - W(\Phi) + \lambda \Phi (\nabla \cdot \mathbf{v}) .} \quad (4)$$

This is the Bernoulli (or generalized Hamilton–Jacobi) relation for the multiplier χ .

4. Variation with respect to the scalar field Φ .

Collect Φ -dependent pieces of the integrand:

$$\mathcal{L} \supset -\rho W(\Phi) - \frac{\kappa}{2}(\nabla\Phi)^2 + \lambda \rho \Phi (\nabla \cdot \mathbf{v}).$$

Take $\Phi \mapsto \Phi + \delta\Phi$. Compute term-by-term.

(i) Potential term:

$$\delta(-\rho W(\Phi)) = -\rho \frac{dW}{d\Phi} \delta\Phi.$$

(ii) Gradient energy:

$$\delta\left(-\frac{\kappa}{2}(\nabla\Phi)^2\right) = -\kappa \nabla\Phi \cdot \nabla(\delta\Phi) = +\kappa \nabla^2\Phi \delta\Phi + (\text{boundary term}),$$

after integrating by parts once and discarding boundary contributions.

(iii) Interaction term:

$$\delta(\lambda \rho \Phi \nabla \cdot \mathbf{v}) = \lambda \rho (\nabla \cdot \mathbf{v}) \delta\Phi.$$

Summing these and enforcing stationarity for arbitrary $\delta\Phi$ yields the scalar Euler–Lagrange equation

$$-\rho \frac{dW}{d\Phi} + \kappa \nabla^2\Phi + \lambda \rho (\nabla \cdot \mathbf{v}) = 0.$$

Divide by ρ (assuming $\rho > 0$) and rearrange to the mass-normalized form used in the main text:

$$\boxed{\kappa \frac{\nabla^2\Phi}{\rho} - \frac{dW}{d\Phi} + \lambda (\nabla \cdot \mathbf{v}) = 0.} \quad (5)$$

Multiplying both sides by ρ gives the compact form

$$\boxed{-\rho \frac{dW}{d\Phi} + \kappa \nabla^2\Phi + \lambda \rho \nabla \cdot \mathbf{v} = 0.}$$

Remark on the scalar’s temporal dynamics: the Lagrangian density (1) contains no explicit time-derivative term for Φ , therefore the EL equation (5) is elliptic/constraint-like rather than a dynamical wave equation. If one wishes to obtain a hyperbolic (wave) evolution for Φ with $\partial_t^2\Phi$ present, introduce a kinetic term in $\mathcal{L}$ of the form

$$\mathcal{L}_{\text{kin},\Phi} = \frac{\mu}{2} (\partial_t\Phi)^2$$

with a phenomenological coefficient μ (with SI units chosen so $\mu(\partial_t\Phi)^2$ is an energy density). Adding that term and redoing the Φ -variation yields

$$\mu \partial_t^2\Phi + \kappa \nabla^2\Phi - \rho \frac{dW}{d\Phi} + \lambda \rho \nabla \cdot \mathbf{v} = 0,$$

which, upon dividing by μ , reproduces the mass-normalized wave-type scalar PDE used earlier (with identification $\mu \leftrightarrow 1$ up to units).

5. From the variational relations to the Eulerian momentum equation.

Equations (3) and (4) are a standard Clebsch-style parametrization. To extract an explicit Eulerian momentum-balance law (the convective form written in the main text), take a material time derivative $D_t \equiv \partial_t + \mathbf{v} \cdot \nabla$ of $\nabla \chi$ and combine with (3)–(4). The steps (sketched) are:

$$\begin{aligned} \mathbf{v} &= \nabla \chi + \frac{\lambda}{\rho} \nabla(\rho \Phi), \quad (\text{from (3)}) \\ D_t \mathbf{v} &= D_t(\nabla \chi) + D_t\left(\frac{\lambda}{\rho} \nabla(\rho \Phi)\right). \end{aligned} \quad (6)$$

Using $\nabla(\partial_t \chi + \mathbf{v} \cdot \nabla \chi)$ and substituting (4) for the combination $\partial_t \chi + \mathbf{v} \cdot \nabla \chi$ yields after algebra the familiar convective momentum balance:

$$\rho(\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{v}) = -\rho \nabla W(\Phi) - \kappa(\nabla^2 \Phi) \nabla \Phi - \nabla P + \lambda \nabla(\rho \Phi),$$

where the pressure term P can arise from including a thermodynamic internal-energy contribution in $\mathcal{L}$ (or as a Lagrange multiplier enforcing incompressibility). The curvature body force appears precisely in the form derived earlier:

$$\mathbf{f}_\kappa = -\kappa(\nabla^2 \Phi) \nabla \Phi.$$

This is the momentum equation quoted in the corrected document.

6. Energy identity (sketch).

Define the total energy functional (for $\gamma = 0$ and with no explicit $\partial_t \Phi$ kinetic term)

$$E_{\text{tot}}[t] = \int_{\Omega} \left(\frac{1}{2} \rho \mathbf{v}^2 + \rho W(\Phi) + \frac{\kappa}{2} (\nabla \Phi)^2 \right) d^3x.$$

Differentiate E_{tot} with respect to time and use the continuity equation, momentum balance and the scalar EL relation to rearrange terms. After integrations by parts and assuming vanishing boundary fluxes one obtains $dE_{\text{tot}}/dt = 0$. (If a damping term or explicit Φ -kinetic dissipation is present, the right-hand side is negative definite and energy decays accordingly.)

Summary of final EL system (compact):

Continuity: $\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0,$

Momentum (convective form): $\rho(\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{v}) = -\nabla P - \rho \nabla W(\Phi) - \kappa(\nabla^2 \Phi) \nabla \Phi + \lambda \nabla(\rho \Phi),$

Scalar EL: $-\rho \frac{dW}{d\Phi} + \kappa \nabla^2 \Phi + \lambda \rho \nabla \cdot \mathbf{v} = 0.$

Remark. The derivation above is fully variational and dimensionally consistent. If instead you prefer the scalar field Φ to be explicitly dynamical (wave equation with second time derivative), add a suitable kinetic term $\frac{\mu}{2}(\partial_t \Phi)^2$ to $\mathcal{L}$. The same variational protocol (with the continuity multiplier) then yields the hyperbolic scalar PDE displayed in the main draft (with μ controlling the inertial scale of Φ).